Arava Leela Naga Venkata Sai Kumar

Full Stack Software Engineer | React.js, Node.js, Python, Google Cloud

Hyderabad, India | +91 9000133390 | saikumararava23@gmail.com

linkedin.com/in/sai-kumar-arava-870505201 | github.com/Saikumararava | portfolio-react-inky-mu.vercel.app

PROFESSIONAL SUMMARY

Full Stack Software Engineer and Google Cloud Certified Associate Cloud Engineer experienced in React.js, Node.js, and Python APIs. Built authentication workflows, database integrations, cloud deployments, and automated tests at SentraAuth; contributed to machine learning pipelines and Windows endpoint software.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C#, SQL, HTML5, CSS3.
Web & Databases: React.js, Node.js, Express.js, FastAPI, REST APIs, MongoDB, Mongoose, Firestore, Material UI.
Cloud & DevOps: Google Cloud Platform (GCP), Cloud Run, Cloud Storage, Cloud Build, Terraform, Docker, Git, GitHub Actions, continuous integration and delivery (CI/CD).
Security & Testing: Firebase Authentication, JSON Web Tokens (JWT), role-based access control (RBAC), Identity and Access Management (IAM), pytest, Vitest, Supertest, OpenAPI/Swagger, Postman.
ML & Systems: TensorFlow, PyTorch, PyTorch Lightning, NumPy, Weights & Biases, .NET, Windows Services.

WORK EXPERIENCE

Software Engineer – AI/Cloud & Full Stack | SentraAuth **CIE at IIIT Hyderabad, India**
Progressed from internship to full-time role *October 2025 – Present*

- Built SentraAuth's public-facing company website (sentraauth.com) using the MERN stack (MongoDB, Express.js, React.js, Node.js), developing responsive React pages and connecting to backend services for data persistence, presenting AI-driven continuous-authentication offerings.
- Contributed to Python-based behavioral-biometric ML repositories using PyTorch, PyTorch Lightning, and TensorFlow/Keras. Worked with Transformer architectures (StrokeNet, Keystroke Identity Encoder), LSTM-based Enhanced-TypeNet, DualBranch GPP pipelines, and Hu-CNN models. Handled preprocessing, triplet-loss training, GPU execution, checkpoint management, and tracking via Weights & Biases.
- Developed and maintained a C#/.NET-based Windows agent, handling Windows service lifecycle management, session handling, crash recovery, installer sequencing, and access-control mechanisms.
- Conducted unit and automated testing using Python `pytest` for ML pipelines and APIs alongside manual validation of Windows agent startup, recovery, and CI/CD regression checks.
- Trained models via GCP Workbench, managed model artifacts in Cloud Storage, deployed containerized inference services on Cloud Run, and configured Cloud Build/GitHub Actions CI/CD workflows.

Repository contributions: Inference API, authentication demo, ML training/evaluation pipelines, Windows agent, company website.

PROJECTS

HelpDesk Cloud – Multi-Tenant Helpdesk SaaS
React, TypeScript, Express.js, Firebase Authentication, Firestore, Vitest, Supertest

- Built tenant provisioning, user management, ticket assignment, comments, and status workflows; enforced RBAC and tenant-scoped database access to prevent unauthorized access across organizations.
- Implemented authorized attachments, audit trails, and request tracing. Recorded 88 passing backend tests and 100% coverage of configured source scope; verified attachment storage using the Firebase Storage Emulator.

AI Chatbot – MERN Application GitHub
React, TypeScript, Node.js, Express.js, MongoDB, OpenAI API, Material UI

- Developed a MERN chatbot prototype with a React interface, Express REST API, OpenAI API integration, and MongoDB persistence for user records and assistant responses.
- Implemented signup, login, logout, password hashing, JWT authentication with HTTP-only cookies, and input-validation middleware.

EDUCATION

Bachelor of Technology in Computer Science Engineering **2020 – 2024**
KL University | CGPA: 8.44/10

CERTIFICATIONS

Google Cloud Certified – Associate Cloud Engineer May 2026 – May 2029
Microsoft Certified – Azure Fundamentals 2023
ServiceNow – Certified System Administrator Earned 2023